As provocative and provisionally successful as the old quantum theory was, nobody was really happy with it. Planck and Einstein's idea of light quanta helped make sense of a number of experimental results, but was hard to reconcile with the enormous success of Maxwell's theory of light as electromagnetic waves. Bohr's idea of quantized electron orbits helped make sense of the light emitted and absorbed by hydrogen, but seemed to be pulled out of a hat, and didn't really work for elements other than hydrogen. Even before the "old quantum theory" was given that name, it seemed clear that these were just hints at something much deeper going on.

One of the least satisfying features of Bohr's model was the suggestion that electrons could "jump" from one orbit to another. If a low-energy electron absorbed light with a certain amount of energy, it makes sense that it would have to jump up to another orbit with just the right amount of additional energy. But when an electron in a high-energy orbit emitted light to jump down, it seemed to have a choice about exactly how far down to go, which lower orbit to end up in. What made that choice? Rutherford himself worried about this in a letter to Bohr:

> There appears to me one grave difficulty in your hypothesis, which I have no doubt you fully realize, namely, how does an electron decide what frequency it is going to vibrate at when it passes from one stationary state to the other? It seems to me that you would have to assume that the electron knows beforehand where it is going to stop.

This business about electrons "deciding" where to go foreshadowed a much more drastic break with the paradigm of classical physics than physicists in 1913 were prepared to contemplate. In Newtonian mechanics one could imagine a Laplace demon that could predict, at least in principle, the entire future history of the world from its present state. At this point in the development of quantum mechanics, nobody was really confronting the prospect that this picture would have to be completely discarded.

It took more than ten years for a more complete framework, the "new quantum theory," to finally come on the scene. In fact, two competing ideas were proposed at the time, matrix mechanics and wave mechanics, before they were ultimately shown to be mathematically equivalent versions of the same thing, which can now simply be called quantum mechanics.

Matrix mechanics was formulated initially by Werner Heisenberg, who had worked with Niels Bohr in Copenhagen. These two men, along with